

Proof of Correctness of Prim's Algorithm

- Adding an edge not in a spanning tree creates exactly one cycle
- Removing any edge from this cycle restores a spanning tree

This property is key to our proof

Proof Strategy: Induction

- Let T be the tree built by Prim's algorithm
- Induction Hypothesis: **T is a subgraph of some MST M**
- Base Case: $T = \text{single node} \Rightarrow$ trivially part of any MST
- Inductive Step: Show adding edge e keeps T within some MST

Case 1: Edge Already in MST

- If $e \in M$, then $T \cup \{e\} \subseteq M$
- Inductive hypothesis holds

Case 2: Edge Not in M

- Add e to $M \Rightarrow$ Creates a cycle
- There exists another edge e' in the cycle with:
- e' connects T to outside T
- Prim chose e over e' , so
- $w(e) \leq w(e')$

Swapping Edges

- Replace e' with e in M to get a new tree M'
- Weight of $M' \leq$ Weight of M
- M' is also an MST and M' contains $T \cup \{e\}$

Kruskal's Algorithm (Proof)

- Kruskal's algorithm always maintains a collection of trees (Forest - F)

Proof by induction:

F is always a subgraph of some MST

Kruskal's Algorithm (Proof)

- Base case is true, since F is empty, and empty set is subset of any set

Kruskal's Algorithm (Proof)

- Assume:
F is currently a subgraph of some MST M, and
Kruskal's algorithm adds an edge e
- We want to show that there is some MST M' such that $F \cup \{e\}$ is a subgraph of M0

Kruskal's Algorithm (Proof)

- If $e \in M$ then this is trivially true:
we set $M' = M$, and since $F \subseteq M$ and $e \in M$
we know that $F \cup \{e\} \subseteq M$

Kruskal's Algorithm (Proof)

- There must be some other edge e' whose two endpoints also lie in different trees of F (not necessarily the same two trees as e).
- This means that Kruskal's algorithm could have added e' , but instead chose to add e . Thus $w(e) \leq w(e')$.
- So if we remove e' from M and add e to get a new tree M' , we know that the weight of M' is at most the weight of M and is thus an MST,
- And $F \cup \{e\}$ is a subgraph of M' . This maintains the induction